

Uniform eventual decrease of permanents under matrix powers

A proof of Foregger’s 1978 per- k conjecture

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Abstract

For every n , there is an integer $N(n) \geq 2$ such that $\text{per}(A^k) \leq \text{per } A$ for every nonnegative doubly stochastic $n \times n$ matrix A and every integer $k \geq N(n)$. Equality, for these exponents, holds precisely for uniform block-permutation matrices defined below. This proves the uniform-exponent conjecture attributed to Foregger in Minc’s 1978 catalogue, and strengthens it to all sufficiently large exponents. The proof combines a categorical-occupancy bound for row-averaged matrices with a local estimate that remains valid for arbitrarily large powers. A compactness argument then handles the full Birkhoff polytope without a positive lower bound on its nonzero entries. A uniform quadratic stability consequence is also obtained. The global exponent is not made explicit.

Review status. This is an unreviewed mathematical manuscript, not an accepted publication or an externally certified priority claim. The general theorem is proved analytically; the accompanying exact computations are independent implementation-level diagnostics, not a formal-machine proof of the theorem.

1 Target, prior results, and precise claim

Let Ω_n denote the compact set of real nonnegative matrices with every row and every column summing to one. All powers in this manuscript are *ordinary matrix powers*. Write

$$f(A) = \text{per } A = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}.$$

Foregger’s conjecture asks whether, for each n , there exists an integer $k(n) > 1$, independent of A , for which

$$f(A^{k(n)}) \leq f(A) \quad \text{for every } A \in \Omega_n. \tag{1}$$

The conjecture appears as Conjecture 17 in Minc’s catalogue, and is restated in Cheon–Wanless [2] and Zhang [5]. Melnykova [3, Section 4.1] reports Foregger’s original letter to Minc as dated 17 December 1977; the published catalogue is from 1978. The original letter and the original book page were not independently inspected here. The restriction $k > 1$, made explicit in [3], is essential.

Chang proved the $n = 3$ case, with exponent eight; see [4] and the account in [3]. Melnykova’s Theorem 4.10 gives a common sufficiently-large exponent for matrices whose *nonzero* entries are bounded below by a fixed $c > 0$. Its threshold depends on n and c . Her Corollary 4.11 gives a matrix-dependent exponent for every individual matrix. Removing dependence on a lower-entry bound or on A is the issue addressed here. The present proof does not use either partial result

as an unproved uniform assertion. In a dated primary discussion on 15 May 2022, Wanless again identified this general question as open [7]; that statement is not treated as exhaustive 2026 status evidence.

1.1 The exceptional matrices

Partition $[n] = C_1 \sqcup \cdots \sqcup C_s$ into nonempty classes, put $m_a = |C_a|$, and let π be a permutation of the classes satisfying $m_{\pi(a)} = m_a$. Define

$$F_{ij} = \begin{cases} 1/m_a, & i \in C_a, j \in C_{\pi(a)}, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

Let $\mathcal{F}_n$ be the finite collection of matrices obtainable this way. This definition uses a *common partition of row and column labels*; arbitrary independent row and column relabellings do not describe the same condition for matrix powers.

Associated with this partition is the orthogonal averaging projection

$$E_{ij} = \begin{cases} 1/m_a, & i, j \in C_a, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

It satisfies $E^2 = E = E^\top$ and $EF = FE = F$. Also F^k has the form (2) with π replaced by π^k . Consequently,

$$p := f(F) = f(E) = \prod_{a=1}^s \frac{m_a!}{m_a^{m_a}} > 0, \quad f(F^k) = p \quad (k \geq 1). \quad (4)$$

These matrices include permutations, J_n , direct sums of uniform blocks, and periodic cycles of uniform blocks; they are not all idempotent. Here J_m means the $m \times m$ matrix with every entry $1/m$.

Theorem 1.1. *For every positive integer n , there is $N(n) \geq 2$ such that for every $A \in \Omega_n$ and every integer $k \geq N(n)$,*

$$f(A^k) \leq f(A).$$

Equality holds if and only if $A \in \mathcal{F}_n$.

The theorem compares every sufficiently large power with the *original matrix*, not necessarily with the immediately preceding power. It is not a claim that $f(A^k)$ is monotone in k . The proof gives an explicit local cutoff $16n^2$, but does *not* establish that this is a valid global cutoff.

2 A row-averaging bound with a monotone defect

Throughout the proof $\|\cdot\|_F$ is the Frobenius norm and $\|\cdot\|_2$ the Euclidean operator norm. Every $A \in \Omega_n$ is a Euclidean contraction: Jensen's inequality and the column sums give

$$\sum_i \left| \sum_j a_{ij} x_j \right|^2 \leq \sum_{i,j} a_{ij} |x_j|^2 = \sum_j |x_j|^2.$$

The same statement holds for row vectors by transposition.

Fix a partition and its projection E . For a doubly stochastic matrix M with $EM = M$, define

$$V_E(M) := n - \sum_{a=1}^s \sum_{j=1}^n \left(\sum_{i \in C_a} M_{ij} \right)^2. \quad (5)$$

Lemma 2.1 (Categorical occupancy). *For every $M \in \Omega_n$ satisfying $EM = M$,*

$$f(M) \leq p \left(1 - \frac{V_E(M)}{2n^2} \right) \leq p, \quad p = \prod_a \frac{m_a!}{m_a^{m_a}}. \quad (6)$$

For a fixed $A \in \Omega_n$, the sequence $V_E(EA^k)$ is nondecreasing.

Proof. Rows of M agree within each class. Put $y_{aj} = \sum_{i \in C_a} M_{ij}$, so $M_{ij} = y_{aj}/m_a$ for $i \in C_a$. For each column j , independently choose a class a with probability y_{aj} . These are probabilities because $y_{aj} \geq 0$ and $\sum_a y_{aj} = 1$. Let N_a be the number of columns assigned to class a . Its expectation is $\sum_j y_{aj} = m_a$.

An assignment of columns with exactly m_a columns in each class has $\prod_a m_a!$ row bijections contributing to the permanent. All such bijections for this assignment have the same product of matrix entries. Expanding the permanent therefore gives the exact identity

$$\frac{f(M)}{p} = \mathbb{P}\{N_a = m_a \text{ for every } a\}. \quad (7)$$

Independence across columns implies

$$\mathbb{E}\|N - m\|_2^2 = \sum_{a,j} y_{aj}(1 - y_{aj}) = V_E(M).$$

Since both N and m have nonnegative coordinates summing to n , $\|N - m\|_2^2 \leq 2n^2$. It vanishes on the event in (7). Thus its expectation is at most $2n^2$ times the probability of the complementary event, proving (6).

Let u_a be the column indicator of C_a . Since $u_a^\top E = u_a^\top$,

$$V_E(EA^k) = n - \sum_a \|u_a^\top A^k\|_2^2.$$

Right multiplication by A does not increase any of these squared norms. This proves monotonicity. $\square$

3 Perturbations of a uniform block-permutation matrix

Fix $F \in \mathcal{F}_n$ with partition, E , and p as above. In this section $n \geq 2$. For $A \in \Omega_n$, write

$$X = A - F, \quad \varepsilon = \|X\|_F, \quad \delta = \sum_{F_{ij}=0} A_{ij}. \quad (8)$$

The quantity δ is the total mass leaking outside the support of F . Cauchy–Schwarz gives $0 \leq \delta \leq n\varepsilon$.

Lemma 3.1 (Tangent and leakage components). *There is an orthogonal decomposition $X = Y + Z$ such that Y is supported on $C_a \times C_{\pi(a)}$, has zero row and column sums in each such block, and*

$$EY = YE = 0, \quad \varepsilon^2 = \|Y\|_F^2 + \|Z\|_F^2, \quad \|Z\|_F \leq \sum_{i,j} |Z_{ij}| \leq 4\delta. \quad (9)$$

If $\varepsilon \leq 1/n$, then $F + Y \in \Omega_n$.

Proof. Orthogonally project each allowed block of X onto the subspace with zero row and column sums, and put $Y = 0$ elsewhere. In an allowed $m \times m$ block with row sums r_i , column sums c_j , and total t ,

$$Y_{ij} = X_{ij} - \frac{r_i}{m} - \frac{c_j}{m} + \frac{t}{m^2}.$$

This is the orthogonal projection, so the norm identity holds. The zero block sums give $EY = YE = 0$.

Every r_i, c_j is nonpositive: it is minus the mass of A outside the allowed block in the corresponding row or column. In each block,

$$\sum_{i,j} |Z_{ij}| \leq \sum_i |r_i| + \sum_j |c_j| + |t| = 3(-t).$$

Summing $-t$ over allowed blocks gives δ . The entries of $Z = X$ outside the allowed blocks are nonnegative and total δ , proving the bound 4δ . Finally $\|Y\|_F \leq \varepsilon$, so $\varepsilon \leq 1/n$ keeps all allowed entries of $F + Y$ nonnegative; its row and column sums remain one. $\square$

3.1 Permanent derivatives and a lower bound for the starting matrix

Set

$$C = C_n := \frac{n^2(n-2)!}{2}. \quad (10)$$

On Ω_n , each second partial derivative of f has absolute value at most $(n-2)!$: it is either zero or the permanent of a submatrix with entries at most one. Since $\sum |H_{ij}| \leq n \|H\|_F$, the Hessian operator norm is at most $2C$. In particular, for a segment in Ω_n ,

$$|f(M+D) - f(M) - Df_M(D)| \leq C \|D\|_F^2. \quad (11)$$

For $n \geq 3$ the third Taylor remainder in a direction Y on such a segment is bounded by

$$\frac{n^3(n-3)!}{6} \|Y\|_F^3 \leq C \|Y\|_F^3.$$

For $n = 2$ the third remainder is zero.

The permanent cofactor at an allowed position of F equals p , whereas at a position outside its support it is zero. Therefore

$$Df_F(X) = -p\delta, \quad Df_F(Y) = 0, \quad Df_F(Z) = -p\delta. \quad (12)$$

For a tangent block Y_a of size $m_a > 1$, direct expansion at J_{m_a} gives the quadratic coefficient

$$f(F + tY) = p + t^2 p \sum_{a:m_a > 1} \frac{m_a}{2(m_a - 1)} \|Y_a\|_F^2 + O(t^3). \quad (13)$$

To verify the coefficient, the sum of $Y_{ij}Y_{h\ell}$ over distinct rows and columns equals $\|Y_a\|_F^2$, because the row and column sums vanish. Passing from ordered to unordered row pairs introduces $1/2$; the complementary permanent equals $(m_a - 2)!/m_a^{m_a-2}$. Dividing by $m_a!/m_a^{m_a}$ gives the displayed factor. Blocks of size one have no tangent directions, and cross-block linear products vanish.

If $\varepsilon \leq 1/n$ and $C\varepsilon \leq p/4$, the remainder bound shows

$$f(F + Y) \geq p + \frac{p}{4} \|Y\|_F^2. \quad (14)$$

The segment $F + Y + tZ$, $0 \leq t \leq 1$, lies in Ω_n , and its distance from F is at most ε by orthogonality. The Hessian bound and (9) therefore give

$$|f(A) - f(F + Y) - Df_F(Z)| \leq 2C\varepsilon \|Z\|_F \leq 8C\varepsilon\delta.$$

If also $8C\varepsilon \leq p$ and $\delta \leq 1/4$, then

$$\begin{aligned} f(A) &\geq p - 2p\delta + \frac{p}{4} \|Y\|_F^2 \\ &\geq p - 2p\delta + \frac{p}{4} \varepsilon^2 - 4p\delta^2 \\ &\geq p - 3p\delta + \frac{p}{4} \varepsilon^2. \end{aligned} \tag{15}$$

4 An estimate uniform in arbitrarily large powers

The difficulty is that A^k need not stay near F^k when k is large. The following argument differentiates a *fixed* power only, then uses a monotone quantity to cover all later powers.

Lemma 4.1 (Fixed-time growth of the mixing defect). *For an integer $r \geq 1$, set $g_r(A) = V_E(EA^r)$. Then*

$$g_r(F + Y) = 0, \quad Dg_r(F)[X] = 2r\delta. \tag{16}$$

If $\varepsilon \leq 1/n$ and $\varepsilon \leq 1/(16nr)$, then

$$V_E(EA^k) \geq r\delta \quad \text{for every } k \geq r. \tag{17}$$

Proof. The class indicators satisfy $u_a^\top F = u_{\pi(a)}^\top$ and $u_a^\top Y = 0$. Thus $u_a^\top (F + Y)^r = u_{\pi^r(a)}^\top$, proving the first identity in (16).

The derivative of A^r at F is

$$\mathcal{D}_r(X) = \sum_{j=0}^{r-1} F^j X F^{r-1-j}.$$

For each summand, multiplication by the powers of F carries allowed positions of X to the support of F^r and disallowed positions to its complement. This follows by following the class permutation π ; the correspondence on classes is bijective. The multipliers are doubly stochastic, so the total mass carried from disallowed positions is exactly δ . This remains valid when one of the powers is $F^0 = I$. The derivative therefore has outside mass $r\delta$ and inside total $-r\delta$. Differentiating (5) at the deterministic class assignment of F^r gives $Dg_r(F)[X] = 2r\delta$.

For the error estimate, let U have rows $u_a^\top$; then $\|U\|_F^2 = n$ and $g_r(A) = n - \|UA^r\|_F^2$. On Ω_n , the operator-norm bounds for the first and second derivatives of A^r are respectively $r\|H\|_F$ and $r(r-1)\|H\|_F\|K\|_F$. The product rule yields

$$|D^2g_r(A)[H, K]| \leq 4nr^2\|H\|_F\|K\|_F.$$

Comparing along $F + Y + tZ$ and using $g_r(F + Y) = 0$ consequently gives

$$|g_r(A) - Dg_r(F)[Z]| \leq 4nr^2\varepsilon\|Z\|_F \leq 16nr^2\varepsilon\delta.$$

Also $Dg_r(F)[Y] = 0$, so $Dg_r(F)[Z] = 2r\delta$. For $\varepsilon \leq 1/(16nr)$ the error is at most $r\delta$. This proves $g_r(A) \geq r\delta$. Lemma 2.1 makes the same lower bound valid for every later power, proving (17). $\square$

Lemma 4.2 (Uniform suppression of non-averaged rows). *If $\varepsilon \leq 1/2$, then for every integer $k \geq 1$,*

$$\|(I - E)A^k\|_F \leq 8\delta + \varepsilon^k. \quad (18)$$

Moreover,

$$f(A^k) \leq f(EA^k) + C \|(I - E)A^k\|_F^2. \quad (19)$$

Proof. Put $D_k = (I - E)A^k$, $R = (I - E)AE$, and $T = (I - E)A(I - E)$. Since $(I - E)F = F(I - E) = 0$ and $YE = 0$,

$$\|R\|_F = \|(I - E)ZE\|_F \leq 4\delta, \quad \|T\|_2 \leq \varepsilon, \quad \|D_1\|_F \leq \varepsilon.$$

The exact recurrence is $D_k = RA^{k-1} + TD_{k-1}$. Contraction of powers of A gives

$$\|D_k\|_F \leq 4\delta \sum_{j=0}^{k-2} \varepsilon^j + \varepsilon^k \leq 8\delta + \varepsilon^k.$$

For $M = EA^k$, the rows within a class are equal, so their permanent cofactor rows are equal as well. Since $ED_k = 0$, the sum of the rows of D_k within each class is zero; hence $Df_M(D_k) = 0$. Both M and $M + D_k = A^k$ belong to Ω_n . Applying (11) proves (19). $\square$

Theorem 4.3 (Explicit uniform local gap). *For $n \geq 2$, $F \in \mathcal{F}_n$, and its associated p , set*

$$r = 16n^2, \quad C = \frac{n^2(n-2)!}{2}, \quad \rho_F = \min \left\{ \frac{1}{16nr}, \frac{p}{128Cn} \right\}. \quad (20)$$

For every $A \in \Omega_n$ with $\varepsilon = \|A - F\|_F \leq \rho_F$ and every integer $k \geq r$,

$$f(A) - f(A^k) \geq 4p\delta + \frac{p}{8}\varepsilon^2, \quad \delta = \sum_{F_{ij}=0} A_{ij}. \quad (21)$$

In particular the inequality is strict unless $A = F$.

Proof. The radius guarantees all hypotheses used in (15), including $\delta \leq n\varepsilon \leq p/(128C) \leq 1/4$. It also guarantees the hypotheses of Lemmas 4.1 and 4.2. Combining those lemmas with (6) gives, for every $k \geq r$,

$$\begin{aligned} f(A^k) &\leq p - \frac{pr\delta}{2n^2} + C(8\delta + \varepsilon^k)^2 \\ &\leq p - 8p\delta + 128C\delta^2 + 2C\varepsilon^{2k}. \end{aligned}$$

The first error term is at most $p\delta$. Since $\varepsilon \leq p/(128Cn) \leq p/(16C)$, $\varepsilon \leq 1$, and $2k - 2 \geq 1$, the second error term is at most $(p/8)\varepsilon^2$. Therefore

$$f(A^k) \leq p - 7p\delta + \frac{p}{8}\varepsilon^2.$$

Subtract this from (15) to obtain (21). No bound on $k\varepsilon$ or $k\delta$ was imposed. $\square$

5 The global compactness argument

We use the classical van der Waerden–Falikman–Egorychev theorem:

$$f(T) \geq \frac{m!}{m^m} \quad (T \in \Omega_m), \quad \text{with equality exactly when } T = J_m. \quad (22)$$

The lower bound and its equality characterization are existing results, not ingredients claimed as new here; see [6] and its references to Falikman and Egorychev.

Lemma 5.1 (Limits of powers near a fixed matrix). *For any $A_* \in \Omega_n$, there is a block-averaging projection E as in (3) such that:*

- (i) $f(A_*) \geq f(E)$, with equality exactly when $A_* \in \mathcal{F}_n$;
- (ii) if $A_j \rightarrow A_*$, $k_j \rightarrow \infty$, and $A_j^{k_j} \rightarrow B$, with $A_j \in \Omega_n$, then $EB = B$ and consequently $f(B) \leq f(E)$.

Proof. Recall the standard finite Markov-chain cyclic decomposition, also used for this problem in [3]. A doubly stochastic matrix has no transient states: an isolated sink strongly connected component in its directed graph can have no incoming mass, by equality of total row and column sums; remove such components successively. Thus, after a *simultaneous* row/column permutation, A_* is a direct sum of irreducible doubly stochastic matrices.

Within an irreducible component of period d , split into its cyclic classes. Transitions go from each class to the next. The total outgoing mass of one class equals its size, and the incoming mass of the next equals that next class's size. These sizes are therefore all equal. Each transition block T_a between consecutive classes is square and doubly stochastic. The usual convergence theorem for an irreducible aperiodic finite chain, applied to the d -step restrictions, shows that for a common multiple L of the periods,

$$A_*^{Lt} \longrightarrow E \quad (t \rightarrow \infty), \quad (23)$$

where E averages within every cyclic class. This is an orthogonal projection because the stationary distribution is uniform.

Every perfect matching of the support must match each row class to its unique succeeding column class. Hence the permanent factorizes as $\prod_a f(T_a)$. Applying (22) block by block gives $f(A_*) \geq \prod_a m_a! / m_a^{m_a} = f(E)$. Equality forces every $T_a = J_{m_a}$, exactly the definition of $\mathcal{F}_n$.

For the varying-matrix limit, fix an integer t . For all sufficiently large j , $k_j \geq Lt$. Euclidean contraction implies

$$\|(I - E)A_j^{k_j}\|_2 \leq \|(I - E)A_j^{Lt}\|_2.$$

Taking $j \rightarrow \infty$, then $t \rightarrow \infty$ in (23), gives $(I - E)B = 0$. The limit B is doubly stochastic, so Lemma 2.1 gives $f(B) \leq f(E)$. There is no assumption that k_j is divisible by L , and no assumption of convergence uniform over the matrices A_j . $\square$

Proof of Theorem 1.1. For $n = 1$ the assertion is immediate. Suppose for some fixed $n \geq 2$ that no eventual threshold with the claimed strictness exists. Then there are integers $k_j \rightarrow \infty$ and matrices $A_j \in \Omega_n \setminus \mathcal{F}_n$ such that

$$f(A_j^{k_j}) \geq f(A_j).$$

By compactness, pass to a subsequence with $A_j \rightarrow A_*$ and $A_j^{k_j} \rightarrow B$. Lemma 5.1 and continuity give

$$f(A_*) \leq f(B) \leq f(E) \leq f(A_*).$$

All inequalities are equalities. Thus $A_* = F \in \mathcal{F}_n$. For large j , $\|A_j - F\|_F \leq \rho_F$ and $k_j \geq 16n^2$. Since $A_j \neq F$, Theorem 4.3 then gives $f(A_j^{k_j}) < f(A_j)$, a contradiction. The exceptional matrices attain equality for every positive exponent by (4). $\square$

6 A uniform quadratic stability consequence

The same argument provides more than a strict inequality.

Corollary 6.1. *For every n there are $c_n > 0$ and $N_n \geq 2$ such that for every $A \in \Omega_n$ and every integer $k \geq N_n$,*

$$f(A) - f(A^k) \geq c_n \operatorname{dist}_F(A, \mathcal{F}_n)^2. \quad (24)$$

Neither the global c_n nor N_n is made explicit here.

Proof. If this were false for a fixed n , choose $k_j \geq j$ and $A_j \in \Omega_n$ with

$$f(A_j) - f(A_j^{k_j}) < \frac{1}{j} \operatorname{dist}_F(A_j, \mathcal{F}_n)^2.$$

No A_j is exceptional. Pass to compact limits as before. The squared distances are bounded, so the limiting permanent gap is nonpositive. Lemma 5.1 again forces the limit $A_* = F \in \mathcal{F}_n$. For all sufficiently large j , Theorem 4.3 gives

$$f(A_j) - f(A_j^{k_j}) \geq \frac{p}{8} \|A_j - F\|_F^2 \geq \frac{p}{8} \operatorname{dist}_F(A_j, \mathcal{F}_n)^2,$$

contradicting $1/j < p/8$. For $n = 1$ the distance vanishes and the assertion is trivial. $\square$

7 Interpretation and scope

The permanent of a transition matrix A is the probability that n independent walkers, started one at each of the n states, occupy *distinct* states after one step. The corresponding probability after k steps is $f(A^k)$. Theorem 1.1 says that, under double stochasticity, every sufficiently late time has probability no larger than the one-step probability, with a threshold depending only on n .

The stronger assertion with $k = 2$ is already false. Van Name’s published 2022 example [7] is

$$A = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad f(A) = \frac{1}{8} < \frac{9}{64} = f(A^2).$$

This example is included only as an attributed scope calibration, not as a new counterexample. It is checked by direct permutation summation and Ryser inclusion–exclusion in the package.

This is not a uniform mixing-time theorem. For example, $A_\eta = (1 - \eta)I + \eta J_n$ satisfies $A_\eta^k = J_n + (1 - \eta)^k(I - J_n)$, so convergence can be arbitrarily slow as $\eta \downarrow 0$. The monotone defect in Lemma 2.1, rather than a uniform mixing estimate, is what makes the proof uniform.

The theorem applies to all real nonnegative doubly stochastic matrices, including reducible, periodic, singular, irrational, and boundary matrices. It does not extend the assertion to arbitrary row-stochastic, complex, or signed matrices. It does not settle the distinct Foregger–Sinkhorn tie-point problem, whose 2026 counterexample is due to Lavi [8], or the separate convex-mixture inequalities also associated with Foregger’s name.

8 Exact diagnostics and review status

The accompanying Python and C++ implementations independently test the identities underlying the proof. Python uses subset dynamic programming for permanents and dual-number binary

exponentiation for power derivatives. C++ uses Ryser inclusion–exclusion, a sequential recurrence for those derivatives, explicit cofactor sums for the permanent Hessian, and a block-matrix projection rather than row/column-mean subtraction. The occupancy identity is checked by categorical count dynamic programming in Python and direct assignment recursion in C++.

On the common 90 fixtures in dimensions two through seven, the two implementations agree on 4,500 exact rational fields in 1,080 records. They check 450 mixing derivatives, 90 tangent Hessians, 90 occupancy identities and 450 specified powers. The fixture family is explicitly defined; it is *not* an exhaustive enumeration of doubly stochastic matrices. Of the 90 fixtures, 69 are default fixtures that need not lie inside the very conservative radius (20); the other 21 are radius-controlled fixtures. Their 42 late-power checks satisfy the explicit local gap, including exponents $16n^2 + 1$. Seventeen deliberate mutations are rejected by the diagnostic guards. Finite checks neither prove the universal quantifiers nor supply the unspecified global threshold.

All mathematical computations use integers or rational numbers. No floating-point experiment, numerical spectral tolerance, or random confidence statement is a premise of the proof. The initial fixture selection is deterministic with a recorded seed; randomness only chooses diagnostic examples.

The proof and code were produced and audited within one assistant session. There has been no external specialist review, independent institutional replication, or formal proof-assistant verification. The dated primary-source search found no earlier publication of the specific uniform theorem or proof, but cannot exclude unindexed or private concurrent work. The bibliography and access limitations are recorded separately in `sources/prior_art.md`.

References

- [1] H. Minc, *Permanents*, 1978, Conjecture 17. Original attribution and catalogue location checked through [2, 3, 5]; the original page was not independently accessed.
- [2] G.-S. Cheon and I. M. Wanless, An update on Minc’s survey of open problems involving permanents, *Linear Algebra and its Applications* **403** (2005), 314–342. doi:10.1016/j.laa.2005.02.030. Author-hosted PDF inspected, especially p. 326, Conjecture 17.
- [3] K. Melnykova, *Notes on Foregger’s Conjecture*, M.Sc. thesis, University of Manitoba, 2012. Official repository PDF inspected, especially Chapter 4 and Theorem 4.10. Bibliographic year verified from the thesis front matter.
- [4] D. K. Chang, On two permanental conjectures, *Linear and Multilinear Algebra* **26** (1990), 207–213. Results attributed through [2, 3, 5]; full original article not independently inspected.
- [5] F. Zhang, An update on a few permanent conjectures, *Special Matrices* **4** (2016), 305–316. doi:10.1515/spma-2016-0030; arXiv:1608.02844.
- [6] L. Gurvits, Van der Waerden/Schrijver–Valiant like conjectures and stable (aka hyperbolic) homogeneous polynomials: one theorem for all, arXiv:0711.3496, version 2. Introduction states the van der Waerden bound and unique equality case, crediting Falikman and Egorychev; an independent polynomial proof is given.
- [7] J. Van Name (8 May 2022) and I. M. Wanless (15 May 2022), answers to “On permanent of a square of a doubly stochastic matrix,” MathOverflow question 422029. Original, dated author discussion, not a peer-reviewed article. Van Name supplies the $k = 2$ counterexample; Wanless states the distinct uniform-exponent problem as open in his account.
- [8] Y. Lavi, A counterexample to the Foregger–Sinkhorn tie-point conjecture, 13 August 2026, arXiv:2608.13025. A different conjecture; not evidence that the powers problem was settled.